

Student number: _____

2023 Trial Higher School Certificate Examination

Mathematics Advanced

General Instructions

- Reading time - 10 minutes
- Working time - 3 hours
- Write using black or blue pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks – 100 Section I - 10 marks (pages 2- 7)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II - 90 marks (pages 9 - 30)

- Attempt Questions 11- 32
- Allow about 2 hours and 45 minutes for this section

Care has been taken to ensure that this paper is free of errors and that it mirrors the format and style of past HSC papers. The questions have been adapted from various sources, in an attempt to provide students with exposure to a broad range of questions. However, there is no guarantee whatsoever that the HSC examination will have similar content, style or format. This paper is intended only as a trial for the HSC examination or as revision leading to the examination.

Section I**10 marks****Attempt Questions 1 – 10****Allow about 15 minutes for this section**Use the multiple-choice answer sheet for Questions 1- 10

Question 1Which inequality gives the domain of $y = \sqrt{5x - 4}$

(A) $x < \frac{5}{4}$

(B) $x \leq \frac{5}{4}$

(C) $x \leq \frac{4}{5}$

(D) $x \geq \frac{4}{5}$

Question 2

The number of hours worked during a week by casual staff in a reception centre is normally distributed with a mean of 16 hours and a standard deviation of 2.5 hours.

What is the percentage of casual staff working fewer than 21 hours in a week?

(A) 47.5%

(B) 84%

(C) 95%

(D) 97.5%

Question 3What is the amplitude and period of $y=4\sin\pi x$?

(A) amplitude = 4 and period = 1

(B) amplitude = π and period = 2

(C) amplitude = 4 and period = 2

(D) amplitude = 4 and period = π

Question 4

The maximum value of $4 - 8\cos(2x)$ is:

- (A) -4
- (B) 4
- (C) -12
- (D) 12

Question 5

$\int e^{5x} dx =$

- (A) $\frac{1}{5}e^{5x} + c$
- (B) $\frac{1}{4}e^{5x} + c$
- (C) $\frac{1}{5}e^{4x} + c$
- (D) $\frac{1}{4}e^{4x} + c$

Question 6

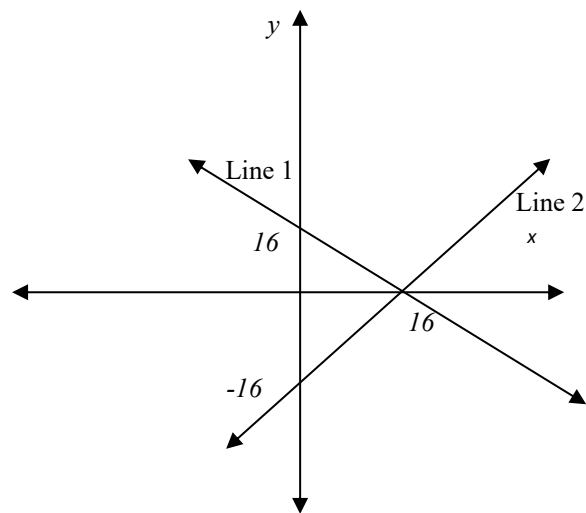
The derivative of $y = \log(\sin x)$ is

- (A) $\tan x$
- (B) $\cot x$
- (C) $\cot^2 x$
- (D) $-\cos x$

Question 7

A computer application was used to draw the graphs of the equations

$$x+y=16 \text{ and } x-y=16$$

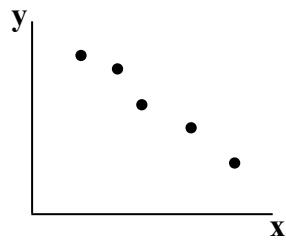


Which row of the table correctly matches the equations with the lines drawn and identifies the solution when the equations are solved simultaneously?

	$x+y=16$	$x-y=16$	Solution
(A)	Line 2	Line 1	$x=16, y=0$
(B)	Line 2	Line 1	$x=16, y=12$
(C)	Line 1	Line 2	$x=16, y=16$
(D)	Line 1	Line 2	$x=16, y=0$

Question 8

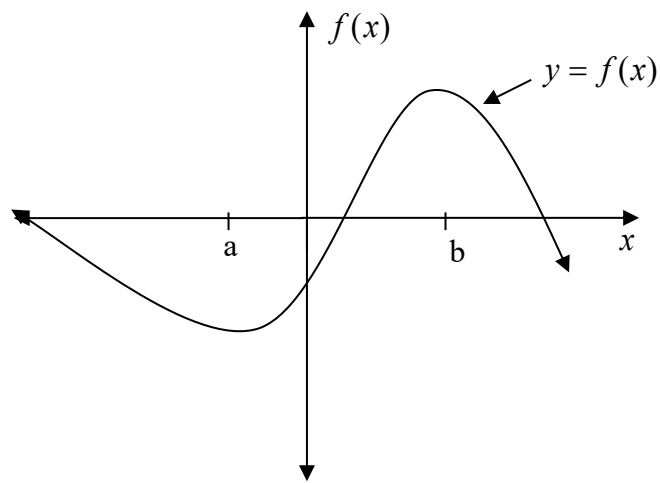
For the data shown in the scatterplot, which value is most likely to be Pearson's correlation coefficient?



- (A) -0.23
- (B) -0.91
- (C) 0.34
- (D) 0.8976

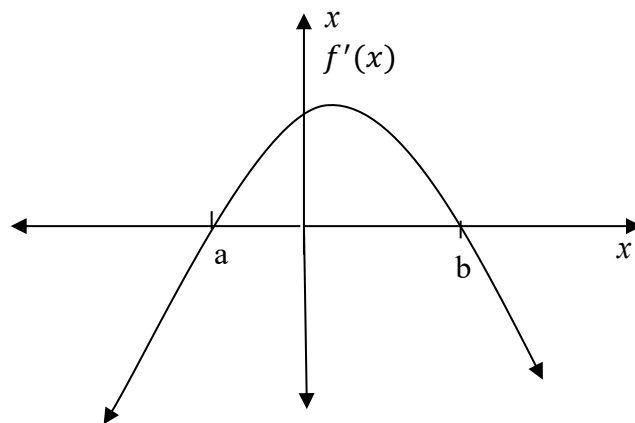
Question 9

The diagram shows the graph $y = f(x)$

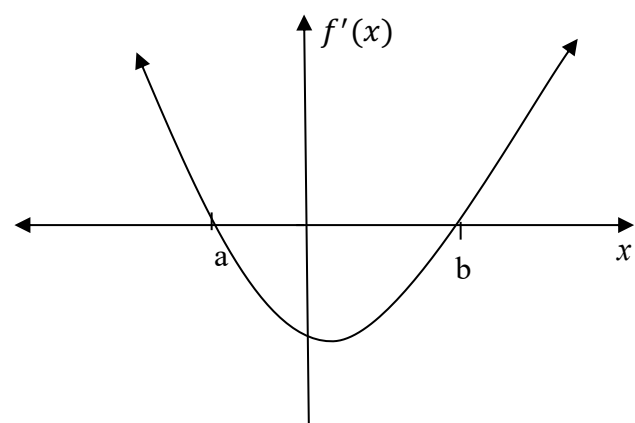


Which of the following graphs shows $y = f'(x)$

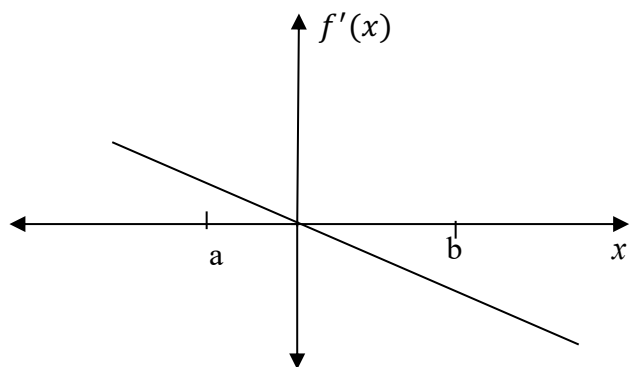
(A)



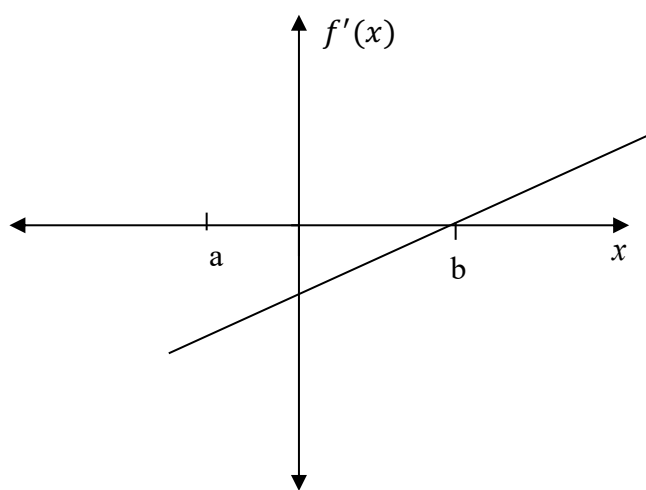
(B)



(C)



(D)



Question 10

The value of x in the equation $\log_2 x - \log_2 (x-2) = \frac{2}{3} \log_2 27$ is

(A) 4

(B) 7

(C) $2\frac{1}{4}$

(D) $3\frac{1}{3}$

**Mathematics Advanced
Section II Answer Booklet**

90 marks

Attempt Questions 11 to 32

Allow about 2 hour and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
 - Your responses should include mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
-

Question 11 (3 marks)

In a school cricket team, the mean weight of players is 62.5 kg and the standard deviation was 3.4 kg. At the same school the mean weight of the soccer team was 67.5 kg and its standard deviation was 3.8.

Karl is in the cricket team and Phil is in the soccer team. Karl weighs 70 kg whilst Phil weighs 75 kg. Which person was heavier in relation to their sporting team?

Question 12 (5 marks)

A company produces 3 tonnes of bread in its first year of production, 7 tonnes in its second year of production and 11 tonnes in its third year of production. The company continues to increase its bread production by 4 tonnes each year thereafter.

- (a) In which year of production did the amount of bread produced for that year, first exceed 85 tonnes?

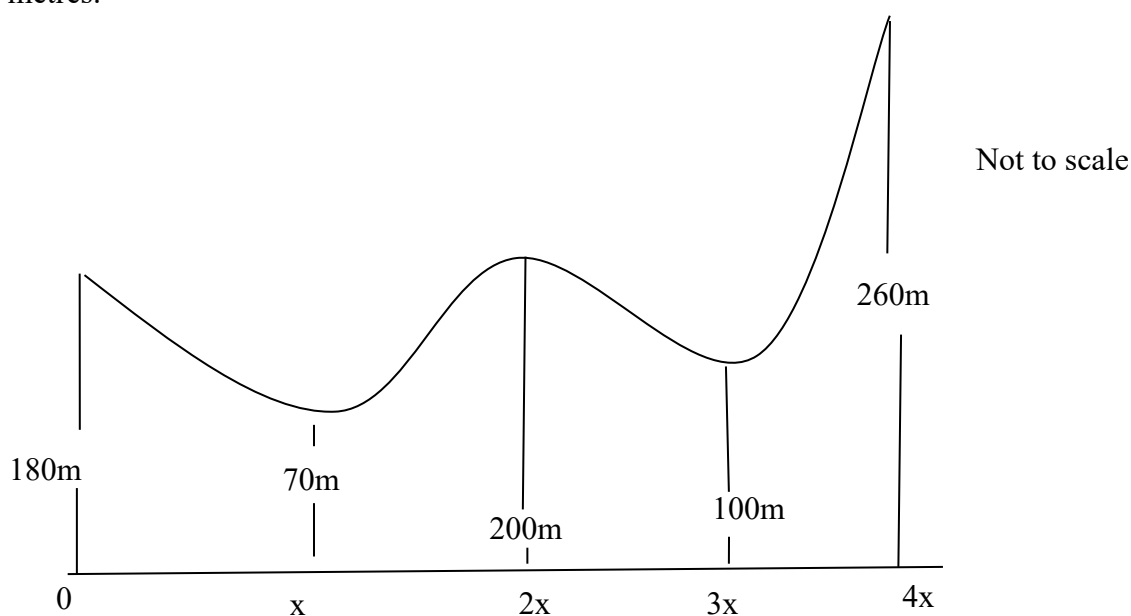
3

- (b) What was the total amount of bread produced by the company in its first 20 years of production?

2

Question 13 (3 marks)

The following diagram represents the depth of a river. The depth of the river is marked every x metres.



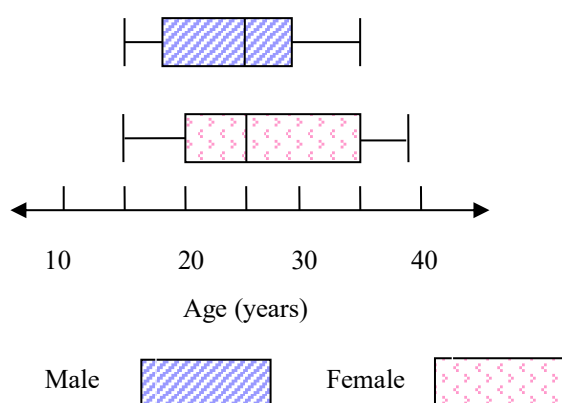
When using trapezoidal's rule with five function values, the area of the cross-section was calculated to be 4130 m^2 . Find the value of x .

Question 14 (2 mark)

The function $y = (x - 3)^3 - 6$ is translated 7 units to the right and 5 units down. Find the equation of the transformed equation.

Question 15 (5 marks)

The box-and whisker plots shown below are based on the ages of cricket champions.



- (a) Write the interquartile range for the ages of the female champions.

1

- (b) Write down the median age of the male champions.

1

- (c) Compare and contrast the displays for male champions and female champions.
Describe the main differences and/or similarities between both sets of data.

3

Question 16 (4 marks)

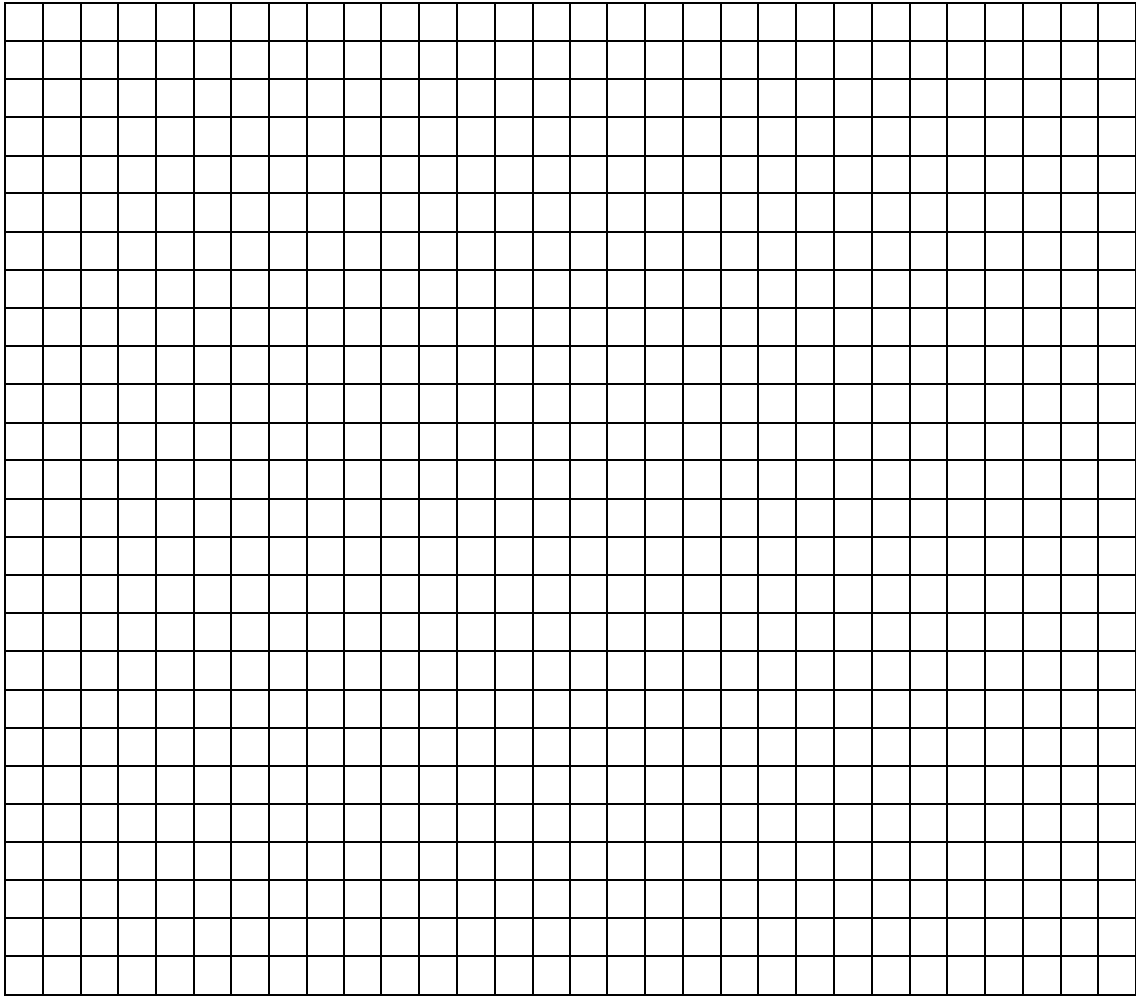
Population (P)	1000	1800	2400	3000	3600	4200	5000
Number of boats stolen (N)	80	78	92	104	104	120	116

The population of a town over a 9 year period and the number of boats stolen each year is recorded above.

- (a)

Graph the data on a scatterplot and draw a line of best fit to represent all data points.
Let the horizontal axis be the population.

3



- (b)

Describe the association between population and number of boats stolen.

1

Question 17 (5 marks)

A survey found that in a large population approximately 25% of people are left-handed.

- (a) Three people are selected at random. Find the probability that at least one of them is left-handed.

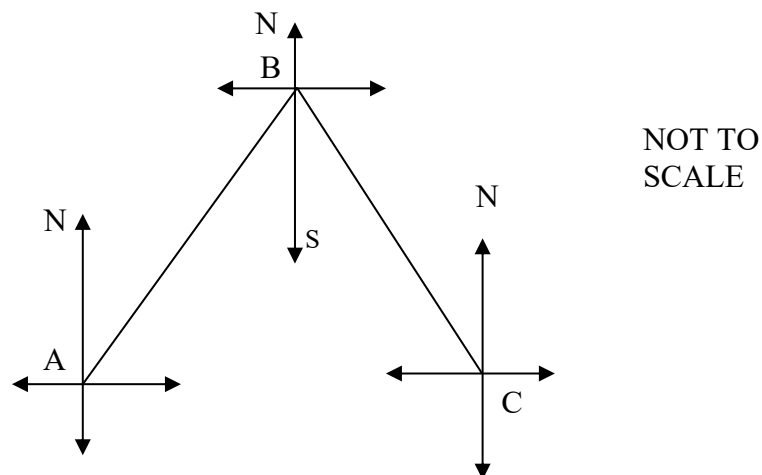
2

- (b) What is the smallest number of people that would need to be selected to have a greater than 99% chance that at least one of them is left-handed.

3

Question 18 (7 marks)

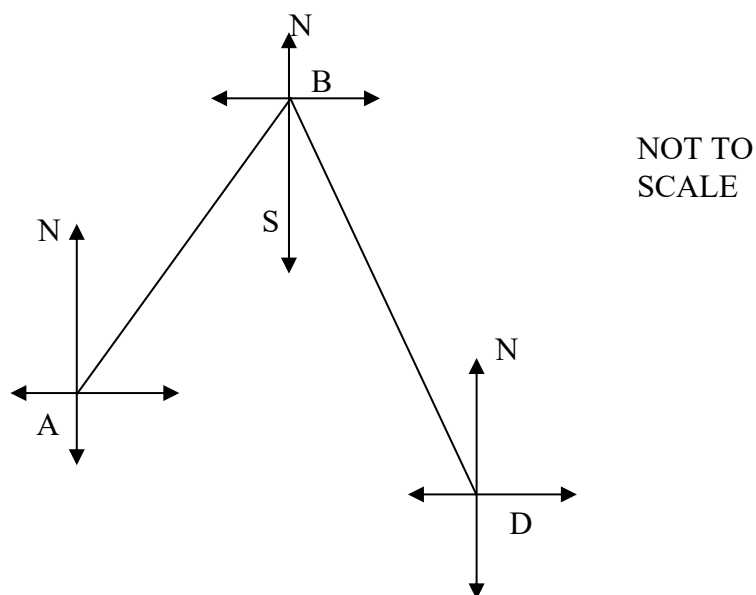
Will runs 20 km from A to B on a bearing of $N38^\circ E$. He then runs from B to C. The bearing of C from B is $S52^\circ E$. C is due east of A.



- (a) Mark the given information on the diagram and explain why $\angle ABC$ is 90° 2

- (b) Find the distance from A to C. Give your answer correct to nearest kilometre. 2

- (c) John runs 20 km from (A) to (B) on a bearing of $N38^\circ E$. He then runs 29 km from B to D. Finally he runs from D to A. The bearing of D from B is $S25^\circ E$.



Find the total distance John ran?

3

Question 19 (3 marks)

Find the exact value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sin x dx$

.

End of Question 19
Questions 11 to 19 are worth 37 marks

Question 20 (5 marks)

(a) Express $\log_9 x$ in the form $k \log_3 x$, where k is a constant

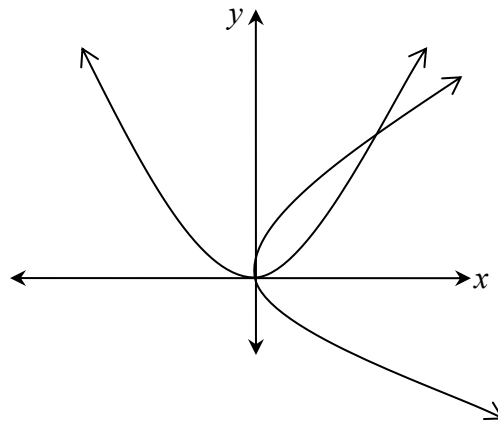
2

(b) Show that the sum to infinity of the series

$$\log_3 x + \log_9 x + \log_{81} x + \dots \text{ is } \log_3 x^2$$

3

Question 21 (5 marks)



The curves $x^2 = 4y$ and $y^2 = 4x$ are sketched.

- (a) Show that the curves intersect at $(0, 0)$ and $(4, 4)$

2

- (b) Find the area enclosed between the two curves.

3

Question 22 (4 marks)

Given $y = xe^{-2x}$, prove that $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$

[illegible]

Question 23 (7 marks)

Jack inherits \$200 000 and invests it in an account earning interest at a rate of 0.5% per month. Each month, immediately after the interest has been paid, Jack withdraws \$2 000.

The amount in the account immediately after the n th withdrawal can be determined using the recurrence relation

$$A_n = A_{n-1}(1.005) - 2000,$$

Where $n=1, 2, 3, \dots$ and $A_0 = 200\,000$

- (a) Use the above recurrence relation to find the amount of money in the account immediately after the third withdrawal. **2**

- (b) Calculate the amount of interest earned in the first three months. **2**

- (c) Calculate the amount of money in the account immediately after the 70th withdrawal. **3**

Question 24 (3 marks)

A discrete random variable X has the probability distribution table shown.

x	1	2	3
$P(X=x)$	0.7	m	0.2

- (a) Find the value of m . **1**

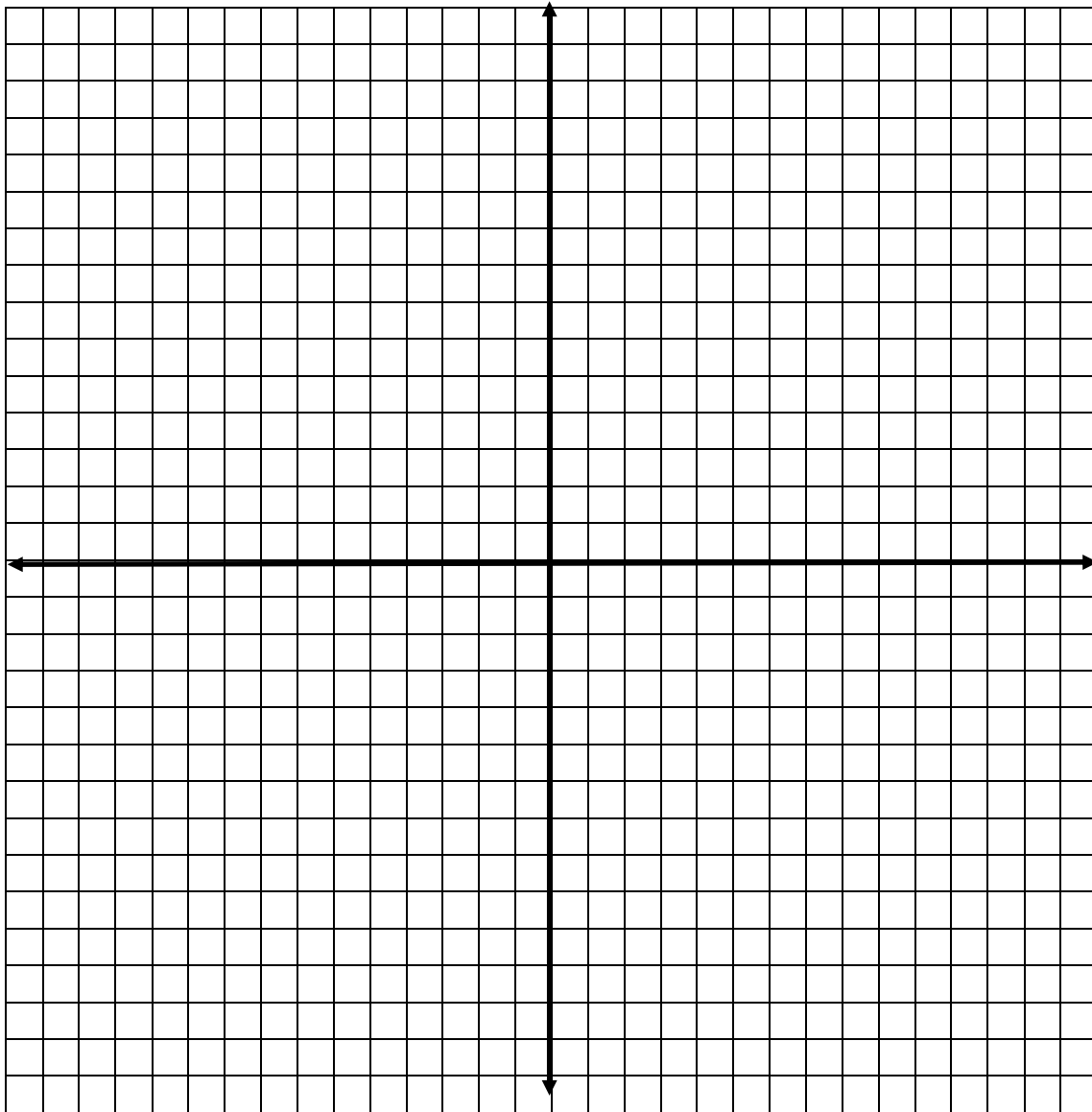
- -----
- (b) Calculate the expected value. **1**

- -----
- (c) Calculate the variance of X . **1**

End of Question 24
Questions 11 to 24 are worth 61 marks

Question 25 (3 marks)

By drawing graphs on the number plane, determine how many solutions there are to the equation $\sin x = \frac{2}{5} \left| \frac{x}{9} \right|$ in the domain $(-\infty, \infty)$



Question 26 (5 marks)

Which option would give the better result at the end of 2 years?

Option A: an annuity of \$2600 a quarter at 4% p.a., compounded quarterly.

Option B: an annuity of \$2800 every six months at 8% p.a. compounded half yearly.

Option C: \$2600 invested at a simple interest rate of 10% p.a.

[illegible]

Question 27 (2 marks)

Use the table of future values of \$1 to find the future value of \$3000 invested per year at 6% p.a. for 9 years, with interest compounded annually.

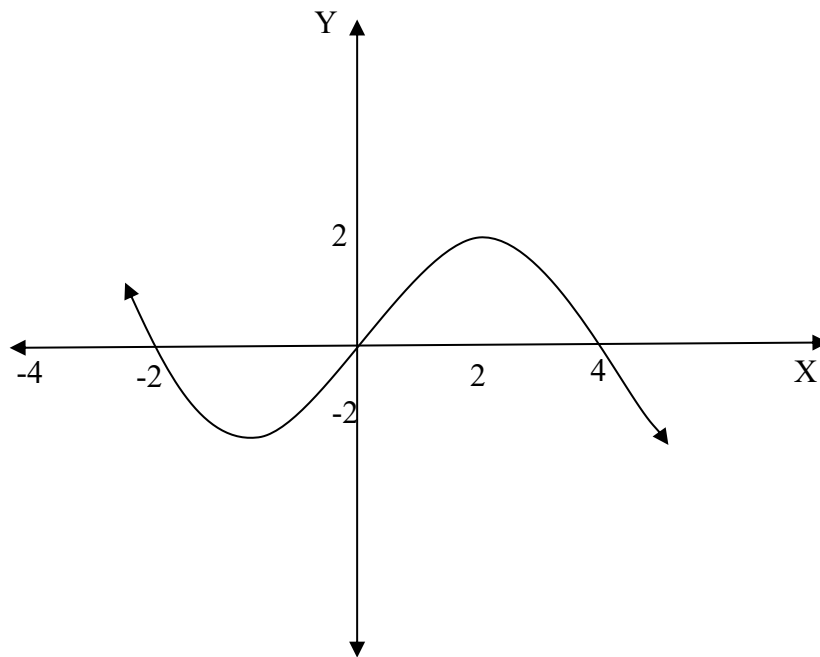
Future values of \$1								
Period	Interest rate (per period)							
	1%	2%	3%	4%	5%	6%	7%	8%
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500	2.0600	2.0700	2.0800
3	3.0301	3.0604	3.0909	3.1216	3.1525	3.1836	3.2149	3.2464
4	4.0604	4.1216	4.1836	4.2465	4.3101	4.3746	4.4399	4.5061
5	5.1010	5.2040	5.3091	5.4163	5.5256	5.6371	5.7507	5.8666
6	6.1520	6.3081	6.4684	6.6330	6.8019	6.9753	7.1533	7.3359
7	7.2135	7.4343	7.6625	7.8983	8.1420	8.3938	8.6540	8.9228
8	8.2857	8.5380	8.8923	9.2142	9.5491	9.8975	10.2598	10.6366
9	9.3685	9.7546	10.1591	10.5828	11.0266	11.4913	11.9780	12.4876
10	10.4622	10.9497	11.4639	12.0061	12.5779	13.1808	13.8164	14.4866

Question 28 (4 marks)

Evaluate $\int_1^3 4x^2\sqrt{x^3 - 1} \, dx$

Question 29 (2 marks)

The graph of $y = f(x)$ is shown.



On the graph above draw the curve $y = f(x - 2) + 2$, showing all important features

Question 30 (2 marks)

Evaluate $\int_2^3 \frac{2x dx}{3x^2 - 4}$

Question 31 (5 marks)

Consider the curve given by $y = 2x^3 - \frac{1}{2}x^4$

- (a) Find the stationary points and determine their nature.

3

(b) Sketch the curve, showing essential features in the domain $[-2, 5]$

2

Question 32 (6 marks)

- (a) A ship sails between two ports. The distance between the ports is B km. The ship has the following costs.

The hourly fuel costs are proportional to the square of the speed. The cost is \$2000 when the speed is 40 knots. All other costs are \$4800 per hour. Let x be the speed in km/h and C the total cost.

Show that $C = B \left(\frac{5}{4}x + \frac{4800}{x} \right)$

3

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[illegible]

End of Exam

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

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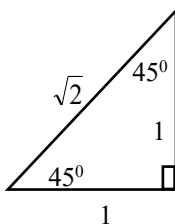
REFERENCE SHEET

Measurement Length $l = \frac{\theta}{360} \times 2\pi r$ Area $A = \frac{\theta}{360} \times \pi r^2$ $A = \frac{h}{2}(a + b)$ Surface area $A = 2\pi r^2 + 2\pi rh$ $A = 4\pi r^2$ Volume $V = \frac{1}{3} Ah$ $V = \frac{4}{3} \pi r^3$	Financial Mathematics $A = P(1 + r)^n$ Sequences and series $T_n = a + (n - 1)d$ $S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$ $T_n = ar^{n-1}$ $S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$ $S = \frac{a}{1 - r}, r < 1$
Functions $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ For $ax^3 + bx^2 + cx + d = 0$: $\alpha + \beta + \gamma = -\frac{b}{a}$ $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$ and $\alpha\beta\gamma = -\frac{d}{a}$ Relations $(x - h)^2 + (y - k)^2 = r^2$	Logarithmic and Exponential Functions $\log_a a^x = x = a^{\log_a x}$ $\log_a x = \frac{\log_b x}{\log_b a}$ $a^x = e^{x \ln a}$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \cos A = \frac{\text{adj}}{\text{hyp}}, \tan A = \frac{\text{opp}}{\text{adj}}$$

$$\text{Area} = \frac{1}{2} ab \sin C$$



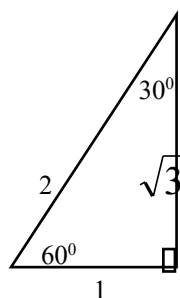
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$\text{Area} = \frac{1}{2} r^2 \theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2} \quad \tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A+B) - \sin(A-B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score

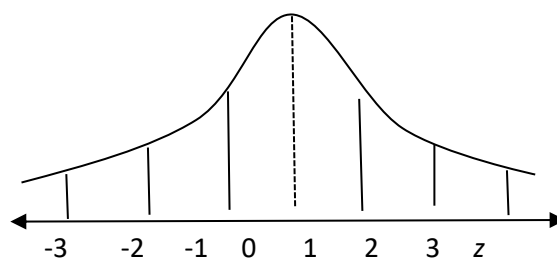
less than $Q_1 - 1.5 \times IQR$

or

more than

$$Q_3 + 1.5 \times IQR$$

Normal distribution



- Approximately 68% of scores have z-scores between -1 and 1
- Approximately 95% of scores have z-scores between -2 and 2
- Approximately 99.7% of scores have z-scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, P(B) \neq 0$$

Continuous random variables

$$P(X \leq x) = \int_a^x f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^n C_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$	
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Differential Calculus		Integral Calculus
Function	Derivative	
$y = f(x)^n$	$\frac{dy}{dx} = nf'(x)[f(x)]^{n-1}$	$\int f'(x)[f(x)]^n dx = \frac{1}{n+1}[f(x)]^{n+1} + c$ where $n \neq -1$
$y = uv$	$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$	$\int f'(x) \sin f(x) dx = -\cos f(x) + c$
$y = g(u)$ where $u = f(x)$	$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$	$\int f'(x) \cos f(x) dx = \sin f(x) + c$
$y = \frac{u}{v}$	$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$	$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$
$y = \sin f(x)$	$\frac{dy}{dx} = f'(x) \cos f(x)$	$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$
$y = \cos f(x)$	$\frac{dy}{dx} = -f'(x) \sin f(x)$	$\int \frac{f'(x)}{f(x)} dx = \ln f(x) + c$
$y = \tan f(x)$	$\frac{dy}{dx} = f'(x) \sec^2 f(x)$	$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$
$y = e^{f(x)}$	$\frac{dy}{dx} = f'(x) e^{f(x)}$	$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$
$y = \ln f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$	$\int \frac{f'(x)}{\sqrt{a^2 + [f(x)]^2}} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$
$y = a^{f(x)}$	$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$	$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$
$y = \log_a f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$	$\int_a^b f(x) dx \approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\}$ where $a = x_0$ and $b = x_n$
$y = \sin^{-1} f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$	
$y = \cos^{-1} f(x)$	$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$	

$y = \tan^{-1} f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$
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Combinations

$${}^n P_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^n C_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1} x^{n-1} a + \dots + \binom{n}{r} x^{n-r} a^r + \dots + a^n$$

Vectors

$$|U| = |xi+yj| = \sqrt{x^2 + y^2}$$

$$u \bullet v = |u| |v| \cos \theta = x_1 x_2 + y_1 y_2$$

Where

$$u = x_1 i + y_1 j$$

$$\text{and } v = x_2 i + y_2 j$$

$$r = a + \lambda b$$

Complex Numbers

$$z = a + ib = r(\cos \theta + i \sin \theta) = r e^{i\theta}$$

$$[r(\cos \theta + i \sin \theta)]^n = r^n (\cos n\theta + i \sin n\theta) = r^n e^{in\theta}$$

Mechanics

$$\frac{d^2 x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

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Centre Number

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Student Number

YEAR 12 Trial HSC EXAMINATION

MATHEMATICS Advanced-MULTIPLE CHOICE ANSWER SHEET

Sample $2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9

(A) ○ (B) ● (C) ○ (D) ○

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) ○ (B) ~~●~~ (C) ● (D) ○

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows

(A) ○ (B) ~~●~~ (C) ~~●~~ (D) ○

correct ↖

- 1 (A) ○ (B) ○ (C) ○ (D) ○
- 2 (A) ○ (B) ○ (C) ○ (D) ○
- 3 (A) ○ (B) ○ (C) ○ (D) ○
- 4 (A) ○ (B) ○ (C) ○ (D) ○
- 5 (A) ○ (B) ○ (C) ○ (D) ○
- 6 (A) ○ (B) ○ (C) ○ (D) ○
- 7 (A) ○ (B) ○ (C) ○ (D) ○
- 8 (A) ○ (B) ○ (C) ○ (D) ○
- 9 (A) ○ (B) ○ (C) ○ (D) ○
- 10 (A) ○ (B) ○ (C) ○ (D) ○

2023

Trial HSC MATHEMATICS ADVANCED

Marking Guidelines

Section I

Multiple-choice Answer key

1	D
2	D
3	C
4	D
5	A
6	B
7	D
8	B
9	A
10	C

1. $5x-4 \geq 0 \therefore x \geq \frac{4}{5}$

2. $\bar{x} + 2\sigma = 16 + 2 \times 2.5 = 21 \therefore 100\% - 2.5\% = 97.5\%$

3. amplitude = 4 and period = $\frac{2\pi}{\pi} = 2$

4. min. value of $\cos(2x)$ is -1 $\therefore$ max. value is $4+8=12$

5. $\int e^{5x} dx = \frac{1}{5} e^{5x} + c$

6. $\frac{dy}{dx} = \frac{\cos x}{\sin x} = \cot x$

7. D

8. B

9. $f'(x) = 0$ at a and b , $f'(x) > 0$ for $a < x < b$, $f'(x) < 0$ for $x < a$ and $x > b$

10.

$$\log_2 x - \log_2 (x-2) = \log_2 27^{2/3} = \log_2 9$$

$$\log_2 \frac{x}{x-2} = \log_2 9$$

$$\frac{x}{x-2} = 9$$

$$x = 9x - 18$$

$$8x = 18$$

$$x = \frac{9}{4} = 2\frac{1}{4}$$

We have endeavored to ensure that the solutions are free of errors and follow the spirit of the syllabus in the methods used to solve the problems. However, individual teachers may opt for alternate solutions and/or may choose a different marking system.

Question 11

Criteria	Marks
• Provides the correct answer	3
• Finds Z score for both Karl and Phil	2
• Finds Z score for either Karl or Phil	1

$$Z_{Karl} = \frac{70 - 62.5}{3.4} = 2.2058$$

$$Z_{Phil} = \frac{75 - 67.5}{3.8} = 1.9736$$

Karl is heavier in relation to his sporting team.

Question 12 (a)

Criteria	Marks
• Provides the correct answer	3
• Provides the majority of the answer	2
• Shows some relevant information	1

$$a=3, d=4$$

$$t_n = 3 + (n-1) \times 4$$

$$4n-1 > 85$$

$$4n > 86$$

$$n > 21.5$$

$$\therefore n = 22 \text{ years}$$

Question 12 (b)

Criteria	Marks
• Provides the correct answer	2
• Shows some relevant information	1

$$s_n = \frac{20}{2} [2 \times 3 + (20-1) \times 4] = 820 \text{ tonnes}$$

Question 13

Criteria	Marks
• Provides the correct answer	3
• Provides the majority of the answer	2
• Provides some relevant information	1

$$4130 = \frac{x}{2} [180 + 260 + 2(70 + 200 + 100)]$$

$$4130 = \frac{1180x}{2} = 590x$$

$$x = 7$$

Question 14

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

$$y = (x - 3 - 7)^3 - 6 - 5$$
$$y = (x - 10)^3 - 11$$

Question 15 (a)

Criteria	Marks
• Provides the correct answer	1

$$35 - 20 = 15$$

Question 15 (b)

Criteria	Marks
• Provides the correct answer	1

25 years

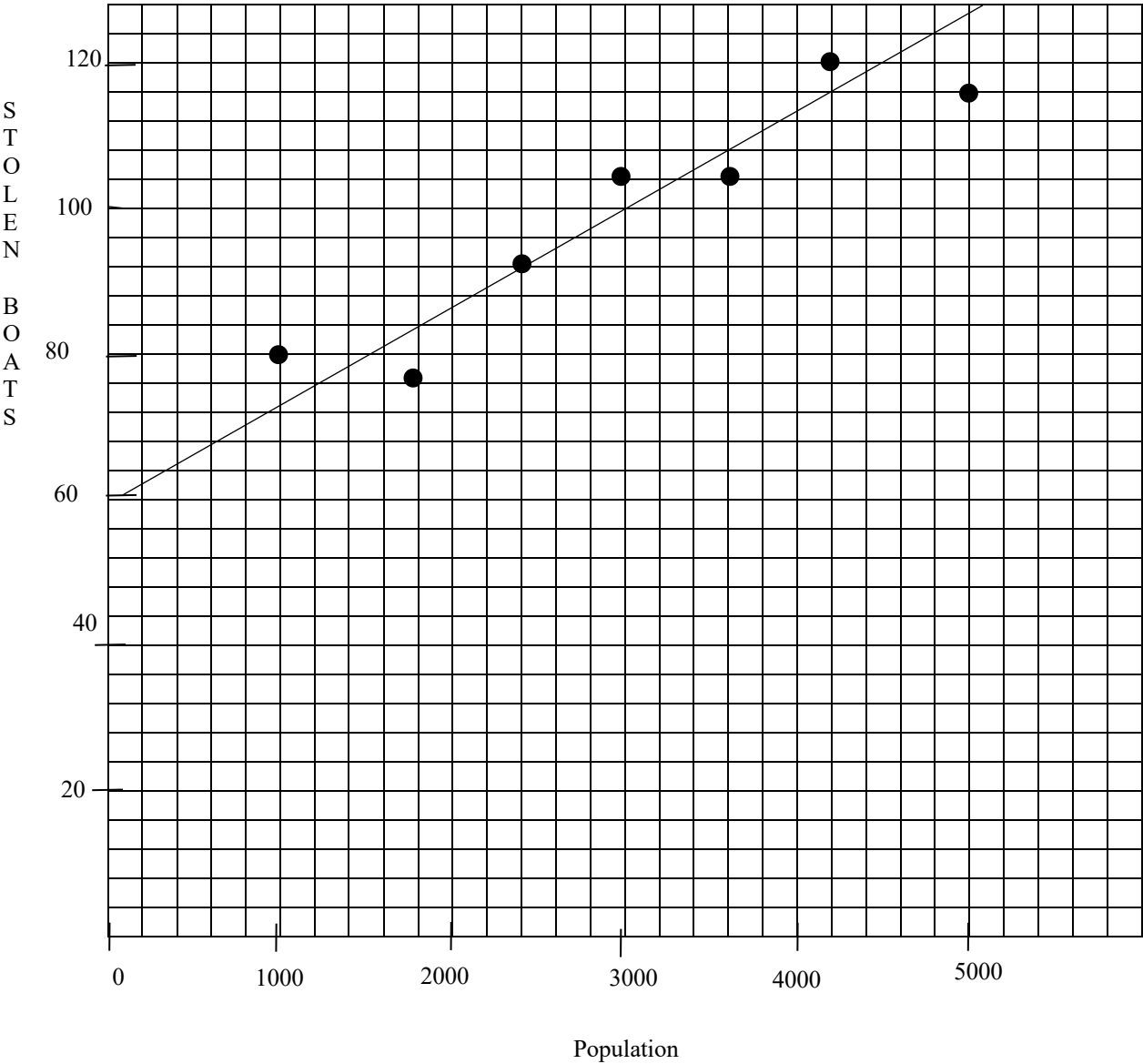
Question 15 (c)

Criteria	Marks
• Provides the correct answer	3
• Provides the majority of the answer	2
• Shows some relevant information	1

Both male and female cricketers have the same median. The female cricketers are positive skewed, whilst the male cricketers are negative skewed. The female cricketers have a larger interquartile range and also a larger range of ages.

Question 16 (a)

Criteria	Marks
• Provides correct solution	3
• Majority of solution is correct	2
• Provides some relevant information	1



Question 16 (b)

Criteria	Marks
• Provides correct solution	1

Strong positive linear association.

Question 17 (a)

Criteria	Marks
• Provides the correct answer	2
• Attempts to use complement, or equivalent merit	1

Let L stand for left-handed and N for not left-handed.

$$P(L)=25\% \quad P(N)=75\%$$

$$\begin{aligned} P(\text{at least one left-handed}) &= 1 - P(NNN) \\ &= 1 - (0.75)^3 = \frac{37}{64} \end{aligned}$$

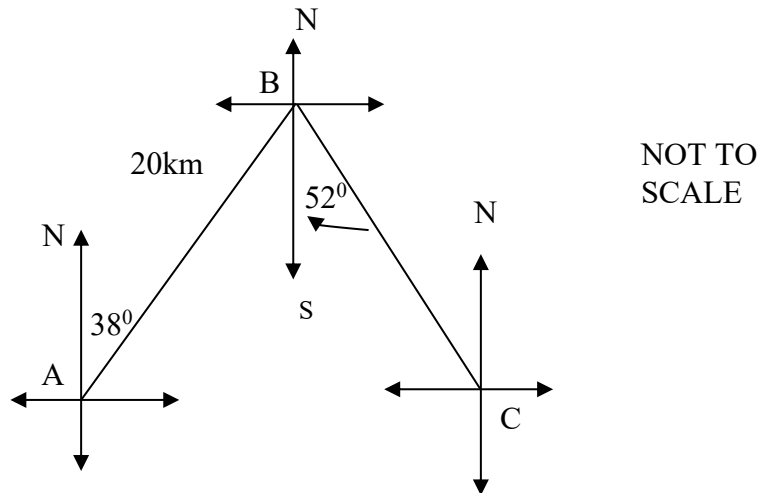
Question 17 (b)

Criteria	Marks
• Provides the correct answer	3
• Finds an inequality in n by taking ln of both sides, or equivalent merit	2
• Provides evidence of expression $1 - (0.75)^n$, or equivalent	1

$$\begin{aligned} 1 - (0.75)^n &> 0.99 \\ 1 - 0.99 &> (0.75)^n \\ 0.01 &> (0.75)^n \\ \ln(0.01) &> n \ln(0.75) \\ n &> \frac{\ln(0.01)}{\ln(0.75)} \\ n &> 16.0078 \\ \therefore &17 \text{ people} \end{aligned}$$

Question 18 (a)

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1



Alternate angles $38^\circ + 52^\circ = 90^\circ$

Question 18(b)

Criteria	Marks
• Provides the correct answer	2
• Provides correct trigonometric ratio	1

$$\cos 52^\circ = \frac{20}{AC}$$

$$AC = \frac{20}{\cos 52^\circ} = 32.485... = 32 \text{ km (nearest km)}$$

Question 18 (c)

Criteria	Marks
• Provides the correct answer	3
• Uses cosine rule correctly	2
• Provides some relevant information	1

$$AD^2 = 20^2 + 29^2 - 2(20)(29)\cos 63^\circ$$

$$= 714.3710205$$

$$AD = 26.73 \text{ km}$$

$$\text{Total distance} = 20 + 29 + 26.73 = 75.73 \text{ km}$$

Question 19

Criteria	Marks
• Provides the correct answer	3
• Majority of the answer is correct	2
• Correct integrand	1

$$\begin{aligned}
 \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sin x dx &= [-\cos x]_{\frac{\pi}{6}}^{\frac{\pi}{4}} \\
 &= \frac{-\sqrt{2}}{2} + \frac{\sqrt{3}}{2} \\
 &= \frac{\sqrt{3}-\sqrt{2}}{2}
 \end{aligned}$$

Question 20 (a)

Criteria	Marks
• Provides the correct answer	2
• Shows some relevant information.	1

$$\begin{aligned}
 \text{let } y &= \log_9 x \\
 \therefore x &= 9^y = 3^{2y} \\
 \text{taking logs to base 3} \\
 \log_3 x &= \log_3 3^{2y} \\
 &= 2y \log_3 3 \\
 y &= \frac{1}{2} \log_3 x
 \end{aligned}$$

Question 20 (b)

Criteria	Marks
• Provides correct answer	3
• Majority of the answer is correct	2
• Provides some relevant information	1

$$\begin{aligned}
 \text{let } y &= \log_{81} x \\
 x &= 81^y = 3^{4y} \\
 \log_3 x &= \log_3 3^{4y} \\
 \therefore y &= \frac{1}{4} \log_3 x \\
 \log_3 x + \log_9 x + \log_{81} x + \dots &= \log_3 x + \frac{1}{2} \log_3 x + \left(\frac{1}{2}\right)^2 \log_3 x + \dots \\
 &= \log_3 x \left[1 + \frac{1}{2} + \frac{1}{4} + \dots \right] \\
 &= \log_3 x \left[\frac{1}{1 - \frac{1}{2}} \right] \\
 &= 2 \log_3 x = \log_3 x^2
 \end{aligned}$$

Question 21(a)

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

$$\begin{aligned}
 x^2 &= 4y \\
 \frac{x^2}{4} &= y \dots\dots A \\
 y^2 &= 4x \dots\dots B \quad \text{sub A into B} \\
 \frac{x^4}{16} &= 4x \therefore x^4 = 64x \\
 x^4 - 64x &= 0 \\
 x(x^3 - 64) &= 0 \\
 \therefore \text{curves intersect at } (0, 0) \text{ and } (4, 4)
 \end{aligned}$$

Question 21(b)

Criteria	Marks
• Provides the correct answer	3
• Integrates correctly	2
• Provides some relevant information	1

$$\begin{aligned}
 A &= \int_0^4 2\sqrt{x} - \frac{x^2}{4} dx \\
 &= \int_0^4 2x^{\frac{1}{2}} - \frac{x^2}{4} dx \\
 &= \left[\frac{4x^{\frac{3}{2}}}{3} - \frac{x^3}{12} \right]_0^4 \\
 &= \frac{32}{3} - \frac{64}{12} = 5\frac{1}{3}
 \end{aligned}$$

Question 22

Criteria	Marks
• Provides the correct answer	4
• Majority of the answer is correct	3
• Finds second derivate	2
• Shows some evidence in using the quotient rule	1

$$\begin{aligned}
 y &= xe^{-2x} \\
 \frac{dy}{dx} &= x(-2e^{-2x}) + e^{-2x}(1) \\
 &= -2xe^{-2x} + e^{-2x} \\
 \frac{d^2y}{dx^2} &= -2x(-2e^{-2x}) + e^{-2x}(-2) - 2e^{-2x} \\
 &= 4xe^{-2x} - 4e^{-2x} \\
 \therefore \frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y &= 4xe^{-2x} - 4e^{-2x} + 4(-2xe^{-2x} + e^{-2x}) + 4xe^{-2x} \\
 &= 4xe^{-2x} - 4e^{-2x} - 8xe^{-2x} + 4e^{-2x} + 4xe^{-2x} \\
 &= 0
 \end{aligned}$$

Question 23 (a)

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

$$\begin{aligned}
 A_0 &= 200\,000 \\
 A_1 &= A_0(1.005) - 2\,000 = 200\,000(1.005) - 2\,000 = 199\,000 \\
 A_2 &= A_1(1.005) - 2\,000 = 199\,000(1.005) - 2\,000 = 197\,995 \\
 A_3 &= A_2(1.005) - 2\,000 = 197\,995(1.005) - 2\,000 = \$196\,984.98
 \end{aligned}$$

Question 23 (b)

Criteria	Marks
• Provides the correct answer	2
• Provides some relevant information	1

$$\begin{aligned}
 200\,000 - 2\,000 \times 3 + I &= 196\,984.98 \\
 194\,000 + I &= 196\,984.98 \therefore I = \$2\,984.98
 \end{aligned}$$

Question 23 (c)

Criteria	Marks
• Provides the correct answer	3
• Makes significant progress	2
• Provides some relevant information	1

$$A_0 = 200\,000$$

$$A_1 = 200\,000(1.005) - 2\,000$$

$$A_2 = [200\,000(1.005) - 2\,000](1.005) - 2\,000$$

$$A_2 = 200\,000(1.005)^2 - 2\,000(1.005) - 2\,000$$

$$A_3 = [200\,000(1.005)^2 - 2\,000(1.005) - 2\,000](1.005) - 2\,000$$

$$A_3 = 200\,000(1.005)^3 - 2\,000(1.005)^2 - 2\,000(1.005) - 2\,000$$

$$A_3 = 200\,000(1.005)^3 - 2\,000(1 + 1.005 + 1.005^2)$$

$$A_n = 200\,000(1.005)^n - 2\,000(1 + 1.005 + 1.005^2 + \dots + 1.005^{n-1})$$

$$A_n = 200\,000(1.005)^n - 2\,000 \times \frac{1.005^n - 1}{0.005}$$

$$A_{70} = 200\,000(1.005)^{70} - 2\,000 \times \frac{1.005^{70} - 1}{0.005} = \$116\,433.89$$

Question 24 (a)

Criteria	Marks
• Provides the correct answer	1

$$\begin{aligned} 0.7 + m + 0.2 &= 1 \\ m &= 0.1 \end{aligned}$$

Question 24 (b)

Criteria	Marks
• Provides the correct answer	1

$$E(x) = 1 \times 0.7 + 2 \times 0.1 + 3 \times 0.2 = 1.5$$

Question 24 (c)

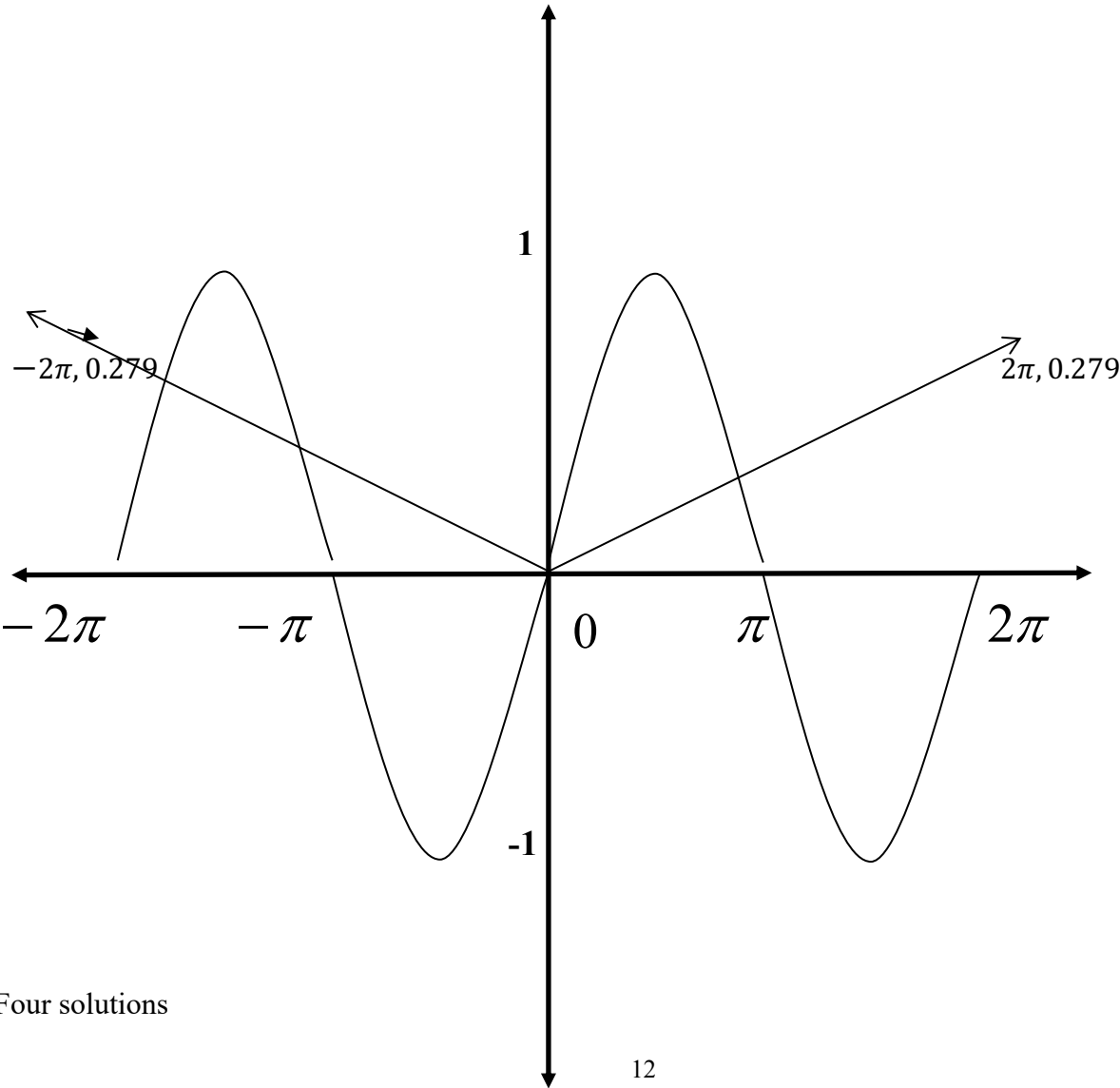
Criteria	Marks
• Provides the correct answer	1

x^2	1	4	9
$P(X=x)$	0.7	0.1	0.2

$E(x^2)=1(0.7)+4(0.1)+9(0.2)=2.9$
 $(E(x))^2=1.5 \times 1.5=2.25$
 $Var(x)=E(x^2)-(E(x))^2=2.9-2.25=0.65$

Question 25

Criteria	Marks
• Provides the correct answer	3
• Draws both graphs	2
• Provides some relevant information	1



Four solutions

Question 26

Criteria	Marks
• Provides the correct answer	5
• Both future values for option A and B are correct	4
• Future value for option A or B is correct	3
• Option C has correct answer	2
• Provides some relevant information	1

Option A:

$$\text{Interest rate} = (4/4)\% = 1\%$$

$$\text{Number of periods} = 2 \times 4 = 8$$

$$\text{Future value} = \$2600 \times (1 + 1/100)^8 = \$2\,815.43$$

Option B:

$$\text{Interest rate} = (8/2)\% = 4\%$$

$$\text{Number of periods} = 2 \times 2 = 4$$

$$\text{Future value} = \$2\,800 \times (1 + 4/100)^4 = \$3\,275.60$$

Option C:

$$\text{Simple interest} = 2\,600 \times 10/100 \times 2 = \$520$$

$$\text{Future value} = 2\,600 + 520 = \$3120$$

Option B is the better result after 2 years.

Question 27

Criteria	Marks
• Provides the correct answer	2
• Provides some information leading to correct answer	1

$$\text{Future value} = 3000(11.4913) = \$34\,473.90$$

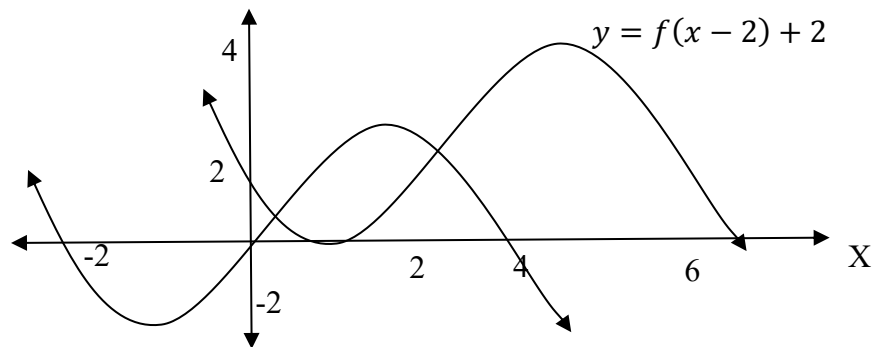
Question 28

Criteria	Marks
• Provides the correct answer	4
• Integrates correctly	3
• Shows $\int_1^3 4x^2\sqrt{x^3-1}dx \left[\frac{1(x^3-1)^{\frac{3}{2}}}{\frac{3}{2}} \right]_1^3$	2
• Provides some information leading to correct answer	1

$$\begin{aligned}
 & \int_1^3 4x^2\sqrt{x^3-1}dx \\
 &= 4 \int_1^3 x^2(x^3-1)^{\frac{1}{2}}dx \\
 &= \frac{4}{3} \int_1^3 3x^2(x^3-1)^{\frac{1}{2}}dx \\
 &= \frac{4}{3} \left[\frac{1(x^3-1)^{\frac{3}{2}}}{\frac{3}{2}} \right]_1^3 \\
 &= \frac{8}{9} \left[\sqrt{(x^3-1)^3} \right]_1^3 \\
 &= \frac{8}{9} (\sqrt{26^3-0}) \\
 &= \frac{8}{9} \sqrt{17576}
 \end{aligned}$$

Question 29

Criteria	Marks
• Provides the correct answer	2
• Draws correct horizontal or vertical transformation	1



Question 30

Criteria	Marks
• Provides the correct answer	2
• Provides some information leading to correct answer	1

$$\begin{aligned}
 & \int_2^3 \frac{2x dx}{3x^2 - 4} \\
 &= \frac{1}{3} \int_2^3 \frac{6x dx}{3x^2 - 4} \\
 &= \frac{1}{3} [\log_e(3x^2 - 4)]_2^3 \\
 &= \frac{1}{3} [\log_e(23) - \log_e(8)] \\
 &= \frac{1}{3} \log_e\left(\frac{23}{8}\right)
 \end{aligned}$$

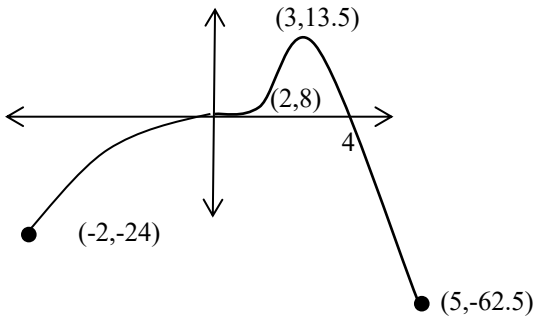
Question 31 (a)

Criteria	Marks
• Provides the correct answer	3
• Finds horizontal point of inflection.	2
• Finds the first derivative	1

$$\begin{aligned}
 y &= 2x^3 - \frac{1}{2}x^4 \\
 \frac{dy}{dx} &= 6x^2 - 2x^3 \\
 &\text{st. pts} \\
 0 &= 6x^2 - 2x^3 \\
 0 &= x^2(3 - x) \\
 \therefore (0, 0) \text{ and } (3, 13.5) \\
 x &\quad -1 \quad 0 \quad 1 \\
 \frac{dy}{dx} &\quad + \quad 0 \quad + \\
 \therefore \text{horizontal point of inflection at } (0, 0) \\
 x &\quad 2 \quad 3 \quad 4 \\
 \frac{dy}{dx} &\quad + \quad 0 \quad - \\
 \therefore \text{rel. max at } (3, 13.5)
 \end{aligned}$$

Question 31 (b)

Criteria	Marks
• Provides the correct answer	2
• Shows correct shape of the curve	1



Question 32 (a)

Criteria	Marks
• Provides the correct answer	3
• Majority of the answer is correct	2
• Makes reasonable progress	1

$$\begin{aligned}
 \text{time} &= \frac{\text{distance}}{\text{speed}} = \frac{B}{x} \\
 \text{hourly cost} &= kx^2 \\
 2000 &= k(40)^2 \\
 k &= \frac{5}{4} \\
 C &= \text{hourly fuel cost} \times \text{number of hours} + \$4800 \text{ per hour} \times \text{number of hours} \\
 &= \frac{5}{4}x^2 \times \frac{B}{x} + 4800 \times \frac{B}{x} \\
 &= B \left(\frac{5}{4}x + \frac{4800}{x} \right)
 \end{aligned}$$

Question 32 (b)

Criteria	Marks
• Provides the correct answer	3
• Majority of the answer is correct	2
• Shows some relevant information	1

$$\begin{aligned}
 C &= B \left(\frac{5}{4}x + \frac{4800}{x} \right) \\
 &= \frac{5}{4}Bx + \frac{4800B}{x} \\
 &= \frac{5}{4}Bx + 4800Bx^{-1} \\
 \frac{dC}{dx} &= \frac{5}{4}B - 4800Bx^{-2} \\
 0 &= \frac{5}{4}B - \frac{4800B}{x^2} \\
 \frac{4800B}{x^2} &= \frac{5}{4}B \\
 4800B &= \frac{5}{4}Bx^2 \\
 x^2 &= 3840 \\
 x &= \sqrt{3840} = 16\sqrt{15} \text{ km/h} \\
 \frac{d^2C}{dx^2} &= \frac{9600}{x^3} \\
 \text{when } x &= 16\sqrt{15} \\
 \frac{d^2C}{dx^2} &> 0 \\
 \therefore \text{economical speed is } 16\sqrt{15} \text{ km/h}
 \end{aligned}$$

Examination Mapping Grid

Section 1

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
1	1	MA-F1 Working with Functions	MA 11-2	1-2
2	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
3	1	MA-T2 Trigonometric Functions and Identities	MA 11-4	3-4
4	1	MA-T2 Trigonometric Functions and Identities	MA 11-4	3-4
5	1	MA-C4 Integral Calculus	MA 12-7	3-4
6	1	MA-C1 Introduction to Differentiation	MA 11-5	2-3
7	1	MA-F1 Working with Functions	MA 11-2	1-2
8	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
9	1	MA-C1 Introduction to Differentiation	MA 11-5	2-3
10	1	MA-C1 Logarithms and Exponentials	MA 11-6	3-4

Section 2

11	3	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
12 (a)	2	MA-M1 Modelling Financial Situations	MA 12-2	3-4
12 (b)	3	MA-M1 Modelling Financial Situations	MA 12-2	3-4
13	3	MA-F1 Working with Functions	MA 11-2	1-2
14	2	MA-F2 Graphing Techniques	MA 12-1	3-4
15 (a)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
15 (b)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
15 (c)	3	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
16 (a)	3	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
16 (b)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
17 (a)	2	MA-S1 Probability and Discrete Probability Distributions	MA 12-8	3-4
17 (b)	3	MA-S1 Probability and Discrete Probability Distributions	MA 12-8	3-4
18 (a)	2	MA-T1 Trigonometry and Measure of Angles	MA 11-3	2-3
18 (b)	2	MA-T1 Trigonometry and Measure of Angles	MA 11-3	2-3
18 (c)	3	MA-T1 Trigonometry and Measure of Angles	MA 11-3	2-3
19	3	MA-C4 Integral Calculus	MA 12-7	3-4
20 (a)	2	MA-M1 Modelling Financial Situations	MA 12-2	3-4
20 (b)	3	MA-M1 Modelling Financial Situations	MA 12-2	3-4
21 (a)	2	MA-C4 Integral Calculus	MA 12-7	3-4
21 (b)	3	MA-C4 Integral Calculus	MA 12-7	3-4
22	4	MA-C3 Applications of Differentiation	MA 12-3	3-4
23 (a)	2	MA-M1 Modelling Financial Situations	MA 12-2	3-4
23 (b)	2	MA-M1 Modelling Financial Situations	MA 12-2	3-4
23 (c)	3	MA-M1 Modelling Financial Situations	MA 12-2	3-4
24 (a)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
24 (b)	1	MA-S2 Descriptive Statistics and Bivariate Data	MA 12-8	2-3

		Analysis		
24 (c)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA 12-8	2-3
25	3	MA-T2 Trigonometric Functions and Identities	MA 11-4	3-4
26	5	MA-M1 Modelling Financial Situations	MA 12-2	3-4
27	2	MA-M1 Modelling Financial Situations	MA 12-2	3-4
28	4	MA-C4 Integral Calculus	MA 12-7	4-5
29	2	MA-F2 Graphing Techniques	MA 12-1	3-4
30	2	MA-C4 Integral Calculus	MA 12-7	4-5
31 (a)	3	MA- C3 Applications of Differentiation	MA 12-3	3-4
31 (b)	2	MA- C3 Applications of Differentiation	MA 12-3	3-4
32 (a)	3	MA-M1 Modelling Financial Situations	MA 12-2	5-6
32 (b)	3	MA- C3 Applications of Differentiation	MA 12-3	5-6